Finite Groups and Their Representations — Exercise Sheet 14

Schur modules, patterns and tableaux

Variant: Questions only

Exercise 1 (From patterns to semistandard tableaux). Let $d \geq 1$ and let λ be a partition with $\ell(\lambda) \leq d$. A semistandard tableau of shape λ has entries in $\{1, \ldots, d\}$, weakly increasing rows and strictly increasing columns. For diagrams $\mu \subseteq \nu$, the *skew diagram* ν/μ consists of the boxes of ν outside μ. It is a *horizontal strip* if no two of its boxes lie in one column. Prove that taking the shapes of the boxes with entries at most k, for $1 \leq k \leq d$, gives a bijection with Lecture 14's GT patterns of top row λ. In particular, prove that the horizontal-strip condition for successive shapes is equivalent to interlacing. Show that the tableau's multiplicity of label k equals the difference of the sums of pattern rows k and $k-1$. Deduce

$$s_\lambda(x_1, \ldots, x_d) = \sum_T x^T$$

from the pattern formula, with x^T the product of the variables named by all entries of T.

Exercise 2 (A Schur character and its weight multiplicities). For $\lambda = (3,2)$ and $d = 3$, enumerate the GT patterns with top row $(3,2,0)$. Compute $s_{(3,2)}(x_1, x_2, x_3)$ in the monomial basis and the dimension of $\mathbb{S}_{(3,2)}(\mathbb{C}^3)$. For $\mu \in \mathbb{Z}^3_{\geq 0}$, the *weight-$\mu$ space* of $\mathbb{S}_{(3,2)}(\mathbb{C}^3)$ consists of the vectors f satisfying

$$\operatorname{diag}(x_1, x_2, x_3) \cdot f = x_1^{\mu_1} x_2^{\mu_2} x_3^{\mu_3} f \qquad \text{for all } x \in (\mathbb{C}^\times)^3.$$

List every pattern of weight $(2,2,1)$ and its semistandard tableau. Explain why the Kostka number $K_{(3,2),(2,2,1)}$ is the dimension of that diagonal weight space, rather than the dimension of the whole Schur module.

Exercise 3 (A new Schur–Weyl decomposition). Decompose $(\mathbb{C}^3)^{\otimes 4}$ under S_4 and $\mathrm{GL}(\mathbb{C}^3)$. Compute all non-zero Schur-module dimensions by GT patterns, and check the total dimension. If A is the image of $\mathbb{C}[S_4]$ on this tensor power, find $\dim A$, $\dim A'$, and the kernel of the map $\mathbb{C}[S_4] \to A$. Keep the dimensions of V^λ and $\mathbb{S}_\lambda(\mathbb{C}^3)$ separate.

Exercise 4 (Symmetric and exterior tensors). For $V = \mathbb{C}^d$, $d \geq 1$, and $n \geq 1$, let

$$P_+ = \frac{1}{n!} \sum_{\sigma \in S_n} \sigma, \qquad P_- = \frac{1}{n!} \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\sigma$$

act by the left place action on $V^{\otimes n}$. Identify their images with $\mathbb{S}_{(n)}(V)$ and $\mathbb{S}_{(1^n)}(V)$, respectively. Define $\mathrm{Sym}^n(V)$ and $\Lambda^n(V)$ here as these symmetric and alternating tensor subspaces. Give explicit bases and prove their dimensions are $\binom{d+n-1}{n}$ and $\binom{d}{n}$, with the latter zero for $n > d$. For two endomorphisms a, b of V, also verify the degree-two polarisation identity expressing $a \otimes b + b \otimes a$ through tensor squares.

Exercise 5 (A Young symmetriser in two different modules). Let V be a complex vector space of dimension $d \geq 1$, and let $n \geq 1$. For a standard tableau T of shape $\lambda \vdash n$, let a_T be the

sum of its row stabiliser and b_T the signed sum of its column stabiliser, both unnormalised. You may use Young's symmetriser theorem in the course's left-ideal convention:

$$\mathbb{C}[S_n]b_T a_T \cong V^\lambda$$

(Fulton–Harris, Theorem 4.3 and Exercise 4.4, with the row/column order converted to this convention). Prove from Wedderburn and Schur–Weyl that the image of $b_T a_T$ on $V^{\otimes n}$ is isomorphic to $\mathbb{S}_\lambda(V)$ as a $\mathrm{GL}(V)$-module. For $T = \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array}$, put $q = b_T a_T$. Expand q in $\mathbb{C}[S_3]$, prove $q^2 = 3q$, and compute the dimensions of $\mathbb{C}[S_3]q$ and $q(V^{\otimes 3})$ for every $d \geq 1$. Explain which group acts on each space.

Exercise 6 (An exterior power of a tensor product). For complex vector spaces A, B of positive dimensions and $n \geq 1$, prove an isomorphism of $\mathrm{GL}(A) \times \mathrm{GL}(B)$-modules

$$\Lambda^n(A \otimes B) \cong \bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(A) \otimes \mathbb{S}_{\lambda'}(B).$$

Terms with $\ell(\lambda) > \dim A$ or $\ell(\lambda') > \dim B$ are zero. Use Schur–Weyl on the two separate tensor powers and take the sign multiplicity for the diagonal S_n-action. You may use Sheet 12's sign-twist result $V^{\lambda'} \cong \mathrm{sgn} \otimes V^\lambda$. You may also use Sheet 6, Exercise 1's equivariant tensor–Hom map $f \otimes w \mapsto (u \mapsto f(u)w)$, which gives $\mathrm{Hom}_G(U, W) \cong (U^* \otimes W)^G$, and $\mathrm{sgn}^* \cong \mathrm{sgn}$. Work out $n = 3$, $\dim A = 2$, $\dim B = 3$, including the dimension sum.

Exercise 7 (⋆ The hook-content formula). For $d \geq 1$ and a partition λ, prove

$$\dim \mathbb{S}_\lambda(\mathbb{C}^d) = \prod_{(i,j)\in\lambda} \frac{d + j - i}{h(i,j)}.$$

For $\ell(\lambda) \leq d$, first prove the equivalent dimension formula

$$s_\lambda(1, \ldots, 1) = \prod_{1 \leq i < j \leq d} \frac{\lambda_i - \lambda_j + j - i}{j - i},$$

with zero padding. Substitute $x_j = \exp(t y_j)$ in the bialternant for pairwise distinct y_j. Expand the determinants at $t = 0$ and compare their first non-zero coefficients. You may take as given the rowwise hook-product identity from Sheet 13: with $N = \ell(\lambda)$,

$$\prod_{(i,j)\in\lambda} h(i,j) = \frac{\prod_{i=1}^N (\lambda_i + N - i)!}{\prod_{1 \leq i < j \leq N} (\lambda_i - \lambda_j + j - i)}.$$

Treat $\ell(\lambda) > d$ separately. Compute the dimension for $\lambda = (3, 2, 1)$ and $d = 4$, and compare with $\dim V^{(3,2,1)}$.